

EVEN SEMESTER EXAMINATION, 2023 – 24
2nd Year/ 4th Semester B.Tech. – Computer Science & Engineering
JAVA PROGRAMMING

Duration: 3:00 hrs

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Explain the following terms using an example with respect to exception handling. i) try ii) catch iii) throw iv) finally b) Describe the concept of object and a class with a suitable Java program? c) Define class Student with suitable data members create two objects using two different constructors of the class? d) Explain the concept of method overriding with an example? How does it facilitate polymorphism? e) Write a program to count number of words from a text file using stream classes? f) Give use of 'this' keyword in Java with suitable example?	5x4=20
Q 2.	Answer any four parts of the following. a) Describe the uses of final and super keywords with respect to inheritance? b) Write a program to copy content of one file into another file? c) Write a program to define class Employee with members as id and salary? Accept data for five employees and display details of employees getting highest salary? d) Differentiate between method overloading and method overriding? e) Explain multithreaded programming in Java. Discuss the thread life cycle and the steps involved in creating threads? f) Describe Applet life cycle with diagram?	5x4=20
Q 3.	Answer any two parts of the following. a) Write a Java program to demonstrate event handling using ActionListener and MouseEvent in AWT? b) What is the purpose of constructors and destructors. Explain in detail with example? c) Write a Java program to implement a simple calculator using classes and methods?	10x2= 20
Q 4.	Answer any two parts of the following. a) Describe the architecture of JDBC (Java Database Connectivity) in Java. How can you execute SQL queries using JDBC? Discuss with example? b) What are the type of lists in XHTML? Create a nested list in XHTML? Modify the XHTML document to make different levels of list in different colors using document style sheet? c) Write a Java program to implement a simple RMI-based client/server application for remote method invocation?	10x2= 20
Q 5.	Answer any two parts of the following. a) How do servlets handle HTTP requests and responses? Provide examples of request and response handling in servlets? b) Write a Java program to create a menu-driven GUI application using AWT? c) Describe in detail three OOPs principles? Illustrate with suitable examples?	10x2= 20